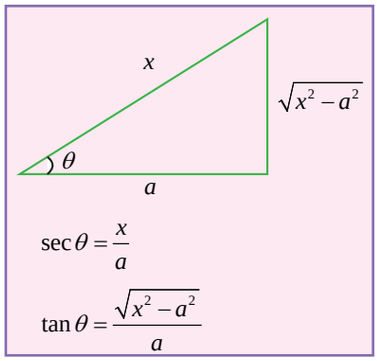
$$= \log\left|\frac{x}{a}+\frac{\sqrt{x^2-a^2}}{a}\right|+c$$

$$= \log\left|x+\sqrt{x^2-a^2}\right|-\log a+c$$

$$= \log\left|x+\sqrt{x^2-a^2}\right|+c_1 \text{ where } c_1 = c-\log a$$

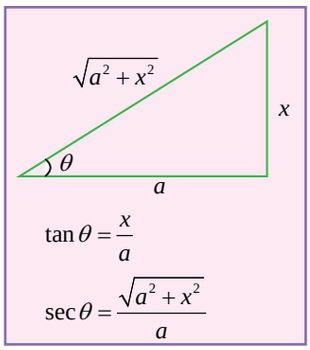
Therefore, $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right|+c_1$



(vi) Let $I = \int \frac{dx}{\sqrt{a^2+x^2}}$

Putting $x = a\ \tan\theta \Rightarrow \theta = \tan^{-1}\left(\frac{x}{a}\right)$

$dx = a\sec^2\theta d\theta$



$$I = \int \frac{a\sec^2\theta}{\sqrt{a^2\tan^2\theta+a^2}}d\theta = \int \frac{a\sec^2\theta}{\sqrt{a^2(\tan^2\theta+1)}}d\theta$$

$$= \int \frac{a\sec^2\theta}{a\sec\theta}d\theta = \int \sec\theta d\theta$$

$$= \log|\sec\theta+\tan\theta|+c$$

$$= \log\left|\frac{x}{a}+\sqrt{\frac{x^2}{a^2}+1}\right|+c$$

$$= \log\left|x+\sqrt{x^2+a^2}\right|-\log a+c$$

$$= \log\left|x+\sqrt{x^2+a^2}\right|+c_1 \text{ where } c_1 = c-\log a$$

$$\int \frac{dx}{\sqrt{a^2+x^2}} = \log\left|x+\sqrt{x^2+a^2}\right|+c_1$$

Remark: Remember the following useful substitution of the given integral as a functions of a^2-x^2, a^2+x^2 and x^2-a^2.

Given	Substitution
a^2-x^2	$x = a\ \sin\theta$
a^2+x^2	$x = a\ \tan\theta$
x^2-a^2	$x = a\ \sec\theta$

11_11th-Maths-Vol2-Unit11.indd 216 16-12-2022 11:07:56